Maaz Ali Mattehalli

Electrical & Electronics Engineer | Energy & Infrastructure Consulting | Techno-Financial Analysis

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Professional Summary

Electrical & Electronics Engineering graduate from NIT Goa with consulting experience in utility infrastructure, techno-financial modelling, regulatory systems, and engineering analytics. Developed decision-support models for government stakeholders, conducted independent EV charging infrastructure diagnostics, and built analytical frameworks supporting infrastructure investment and public-sector transformation. Seeking Energy Market Research and Consulting roles.

Education

National Institute of Technology Goa
B.Tech, Electrical & Electronics Engineering (2026)
Relevant Coursework: Power Systems, Power Electronics, Control Systems, Signals & Systems, Engineering Mathematics

Experience

Product & Innovation Intern — Sep 2025–Present
Athena Infonomics

- Developed an Excel-based techno-financial model supporting investment evaluation for a national water utility project in Lesotho, analysing infrastructure scenarios and ROI for regulatory stakeholders.
- Designed KPI frameworks and performance measurement models for public-sector digital transformation initiatives, translating operational, regulatory, and stakeholder inputs into decision-support tools for infrastructure planning.
- Synthesised evidence from 40+ stakeholder consultations across 4 industry sectors into government-facing deliverables for a MeitY-commissioned national AI adoption study presented at the India AI Summit.
- Standardised research synthesis workflows, reducing analysis turnaround by 60% while improving consistency across consulting deliverables.
- Analysed regulatory systems, operational constraints and implementation risks to support infrastructure and utility-sector recommendations.

R&D Product Intern — Jun 2024 – May 2025
Saafwater

- Led deployment engineering for an industrial IoT water-quality monitoring system, integrating hardware selection, PCB validation, telemetry configuration, and field testing for operational deployment.
- Reduced prototype hardware cost by 75% through structured component evaluation while maintaining ±8% measurement accuracy across calibration trials.
- Prepared engineering documentation covering hardware qualification, testing protocols, calibration records, and deployment procedures for future scale-up.

Selected Impact

- Built ENERNOMICS an engineering decision-support platform to evaluate energy infrastructure investments using financial modelling,technical assumptions, and structured risk assessment.
- Reduced IoT prototype hardware cost by 75% while maintaining ±8% measurement accuracy.
- Built techno-financial model supporting infrastructure investment evaluation for a Lesotho utility project.
- Reduced consulting research turnaround by 60% through structured analytical workflows.
- Identified a 39% telemetry mismatch between charger status and reported availability, highlighting reliability issues affecting charging sessions.
- Revived SPIE chapter, secured US$2,400+ in grants and led initiatives reaching 400+ students.

Selected Work

Enernomics — Decision Intelligence Platform for Energy Infrastructure

- Built an engineering decision-support platform to evaluate energy infrastructure investments using financial modelling, technical assumptions, and structured risk assessment.
- Developed discounted cash flow models incorporating NPV, IRR, payback, and sensitivity analysis to compare investment alternatives.
- Created analytical workflows for EV charging infrastructure and renewable energy procurement, including comparison of Captive, Group Captive, and Third-Party Open Access models.
- Generated executive-ready reports with assumptions, scenario analysis, and supporting visualisations.

EV Charging Infrastructure Diagnostic

- Conducted an independent field study across public EV charging stations in Goa, diagnosing charger-backend synchronisation failures responsible for a 39% telemetry mismatch.
- Quantified operator revenue impact (Rs. 250–400 per failed session) and proposed a phased reliability roadmap incorporating heartbeat verification, monitoring and recovery mechanisms.

Goa EV Charging PPP Financial Model

- Built an Excel-based financial model evaluating utilisation, tariffs, CAPEX, OPEX and ROI for a 25-station EV charging network.
- Performed sensitivity analysis to assess investment feasibility and public-private partnership viability.

Leadership

President, SPIE Student Chapter – Revived dormant chapter, secured US$2,400+ in international grants, organised 6+ workshops reaching 400+ students and established collaborations with IIT chapters.

Skills

Analytics: Advanced Microsoft Excel, Techno-Financial Modelling, Financial Feasibility Analysis, Scenario Analysis, Sensitivity Analysis, ROI Analysis, Root Cause Analysis
Energy: Utility Infrastructure, Energy Transition, EV Charging Infrastructure, Regulatory Systems
Engineering: Industrial IoT, PCB Design, Telemetry, Calibration, Hardware Qualification, Reliability Analysis
Software: Microsoft Office Suite, MATLAB, Simulink, Basic Python, LaTeX